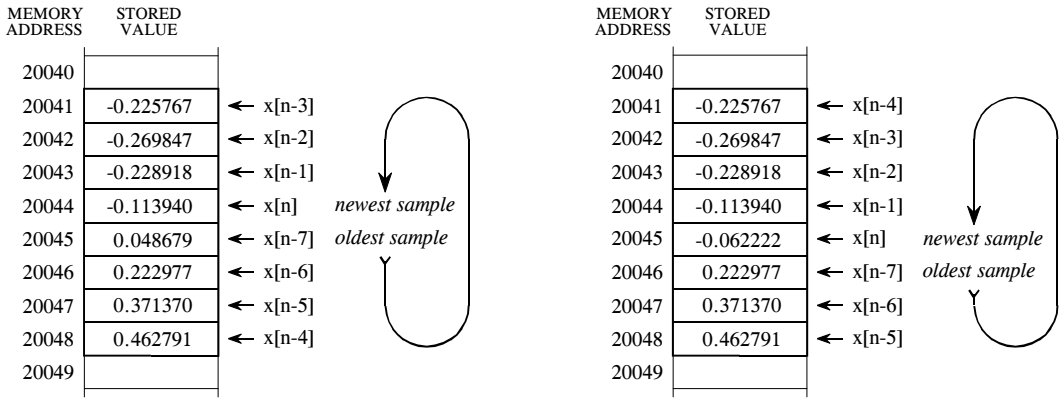

FIGURE 28-3
Circular buffer operation. Circular buffers are used to store the most recent values of a continually updated signal. This illustration shows how an eight sample circular buffer might appear at some instant in time (a), and how it would appear one sample later (b).

of samples, perform the algorithm, and output a group of samples. This is the world of Digital Signal Processors.

Now look back at Fig. 28-2 and imagine that this is an FIR filter being implemented in *real-time*. To calculate the output sample, we must have access to a certain number of the most recent samples from the input. For example, suppose we use eight coefficients in this filter, a_0, a_1, $\cdots a_7$. This means we must know the value of the eight most recent samples from the input signal, $x[n]$, $x[n-1]$, $\cdots x[n-7]$. These eight samples must be stored in memory and continually updated as new samples are acquired. What is the best way to manage these stored samples? The answer is **circular buffering**.

Figure 28-3 illustrates an eight sample circular buffer. We have placed this circular buffer in eight consecutive memory locations, 20041 to 20048. Figure (a) shows how the eight samples from the input might be stored at one particular instant in time, while (b) shows the changes after the next sample is acquired. The idea of circular buffering is that the end of this linear array is connected to its beginning; memory location 20041 is viewed as being next to 20048, just as 20044 is next to 20045. You keep track of the array by a **pointer** (a variable whose value is an *address*) that indicates where the most recent sample resides. For instance, in (a) the pointer contains the address 20044, while in (b) it contains 20045. When a new sample is acquired, it replaces the oldest sample in the array, and the pointer is moved one address ahead. Circular buffers are efficient because only one value needs to be changed when a new sample is acquired.

Four parameters are needed to manage a circular buffer. First, there must be a pointer that indicates the start of the circular buffer in memory (in this example, 20041). Second, there must be a pointer indicating the end of the